b) Three jobs A, B, and C are to be assigned to three machines U, V, and W. The processing cost for each job machine combination is shown in the matrix given below. Determine the allocation that minimizes the overall processing cost. (Cost is in Rs. per unit). **(7)**

JOB	MACHINE		
	U	V	W
A	17	25	31
B	10	25	16
C	12	14	11

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END TERM EXAMINATION

FOURTH SEMESTER [B.TECH] JULY 2023

Paper Code: BS-202 **Subject: Probability, Statistics and Linear Programming**

Time: 3 Hours **Maximum Marks: 75**

Note: Attempt five questions in all including Q.No.1 which is compulsory. Select one question from each unit.

Q1 Attempt all questions:

a) Three urns contain 6 red, 4 black; 4 red, 6 black, and 5 red, 5 black balls respectively. One of the urns is selected and a ball is drawn from it. If the ball drawn is red, find the probability that it is drawn from the first urn. **(3)**

b) A manufacturer produces two types of models M_1 and M_2. Each M_1 model requires 4 hours of grinding and 2 hours of polishing; whereas M_2 model requires 2 hours of grinding and 5 hours of polishing. The manufacturer has 2 grinders and 3 polishers. Each grinder works for 40 hours a week and each polisher works for 60 hours a week. Profit on an M_1 model is ₹ 3 and on an M_2 model is ₹ 4. Whatever is produced in a week is sold in the market. How should the manufacturer allocate his production capacity to the two types of models so that he may make the maximum profit in a week. Formulate this as a linear programming problem. **(3)**

c) Determine the value of c that makes the function $f(x,y) = c(x+y)$, a joint probability mass function over points $x = 1,2,3$ and $y = 1,2,3$. Find $E(X)$ and $E(Y)$. **(3)**

d) Find the line of regression or line of best fit for the data given below: $n = 18, \sum x = 12, \sum y = 18, \sum x^2 = 60, \sum y^2 = 96, \sum xy = 48$. **(3)**

e) Find the moment generating function for the distribution defined by $f(x) = \frac{1}{2^x}$, $x = 1,2,3,\ldots\infty$ **(3)**

UNIT-I

Q2 a) Show that Poisson distribution is a limiting case of Binomial distribution when n is very large and p is small such that np is fixed. If X is a Poisson variate such that $P(X=2) = 9\,P(X=4) + 90\,P(X=6)$. Find the mean and standard deviation of X. **(8)**

b) In the normal distribution, 31% of the items are under 45 and 8% are over 64. Determine the mean and variance of the distribution? (Given that $P(Z < -1.4) = 0.08$ and $P(Z > 0.5) = 0.31$). **(7)**

P.T.O.